

Homework 14: General Knowledge Sweep II — Stretch Chemistry & Reasoning

Paper track. The hardest short-answer content, taken from the toughest reference tests (the SSSS invitational series). Not every test goes this deep — but the harder ones do, and these are worth knowing. No home lab.

Part A: Acids, bases, and bonding (2 pts each)

1. In your own words, what is an **acid** and what is a **base** (either the "donates/accepts H⁺" definition or the "produces H⁺/OH⁻ in water" one)?
2. When an acid dissolves in water, what two things does it form?
3. What is the relationship between pH and hydrogen-ion concentration? (Say whether it's direct or inverse.)
4. Explain the difference between an **ionic bond** and a **covalent bond**.
5. Water is a covalent molecule, but it has a slightly positive end and a slightly negative end. Why? Which atom is which?
6. What is the "octet rule"?

Part B: The starch-iodine test, deeper (2 pts each)

1. Which part of starch — amylose or amylopectin — gives the strong blue-black color, and why?
2. Is the color change a chemical reaction? What is the correct term for what happens?
3. What happens to the color when you heat the mixture, and why?

Part C: DNA & gels (2 pts each)

1. Why is it harder for longer DNA fragments to travel through the gel?
2. Roughly describe how gel electrophoresis works: the well, the gel, the current, and which way the DNA moves.
3. What does CODIS stand for, and what kind of database is it?

Part D: Soil (2 pts each)

1. Give two differences between **peat** and **loam**.
2. What does a **red** soil color indicate? A **gray** color? A **pale/white** color?
3. A soil sample is 55% sand, 20% silt, 25% clay. What is its texture class?
4. Why is soil considered *class* evidence rather than *individual* evidence?

Part E: Glass & spatter reasoning (2 pts each)

1. A window has two bullet holes. Radial cracks from hole 2 stop where they meet cracks from hole 1. Which shot came first, and how do you know?
2. A bullet hole in glass is narrow on the outside and wide on the inside. From which side was the shot fired?
3. Blood stains on a wall all have their long axes pointing back toward one spot on the floor. What is that spot called? What else would you need to know to find the height the blood came from?

Part F: Mixed identification (3 pts each)

For each, name the substance and the one or two tests that clinch it:

1. A white powder: insoluble in water, fizzes with acid, iodine-negative, and turns a "mustard" color with iodine reagent on the grains.
2. A clear liquid: pH 7, faint alcohol smell, evaporates quickly and is flammable.
3. A gray metal: not magnetic, density about 7.1 g/cm³, reacts steadily with HCl, atomic number 30.
4. A plastic: floats in water *and* floats in vegetable oil (0.92), but sinks in 70% isopropyl alcohol (0.79); the burn card says "burns slowly, drips flaming, faint sweet/diesel smell."

Part G: Full analysis practice (10 pts)

Scenario. A rare coin is stolen from a locked display case at a small museum during a fundraiser. The case's glass panel was broken. Evidence: on the broken glass edge, the **radial cracks meet the outer surface at a right angle** (force came from inside the room). A **navy wool fiber** is caught on the case latch. A smear of **white powder** on the case fizzes with acid but is only partly soluble. **Sandy-loam soil** is on the carpet by the case. A **loop fingerprint** is on the latch.

Suspects: **Morgan** (museum janitor, had keys, wears a navy wool coat, no alibi, keeps garden lime by the loading dock where the soil is sandy loam, fingerprints loop/loop). **Riley** (caterer, wears a white cotton apron, was in the kitchen with three witnesses the whole night, fingerprints whorl/loop). **Casey** (donor with a known grudge against the museum director, wears a polyester suit, arrived and left with a group, fingerprints arch/tented arch).

Write a full analysis: culprit, motive, evidence with reasons, timeline, and why each other suspect is cleared.

Answer Key

Part A:

1. Acid = donates H^+ (produces H^+/H_3O^+ in water); base = accepts H^+ (produces OH^- in water).
2. An H^+ (hydrogen) ion and a conjugate base (the rest of the acid molecule, negatively charged).
3. Inverse — as $[H^+]$ goes up, pH goes down ($pH = -\log[H^+]$).
4. Ionic bond: one atom transfers an electron to another, making two oppositely charged ions that attract. Covalent bond: two atoms share a pair of electrons.
5. Oxygen pulls the shared electrons more strongly than hydrogen (electronegativity), so the oxygen end is slightly negative and the hydrogen ends are slightly positive.
6. Atoms tend to gain, lose, or share electrons to end up with eight electrons in their outer shell (two for hydrogen).

Part B:

1. Amylose — its long, straight chains coil into a helix that can hold a long chain of iodine; amylopectin's short branches can't, so it only gives a weak red-brown.
2. No — it's an inclusion complex (a clathrate); the iodine is physically trapped in the helix, forming a charge-transfer complex that absorbs light.
3. The color fades — heat uncoils the helix and the iodine falls out; it returns when the mixture cools and the helix re-forms.

Part C:

1. Larger fragments are bulkier and get caught in the gel's mesh of pores, so they move slowly; small fragments slip through and travel farther.
2. DNA samples are loaded into wells at one end of a gel slab. A current is applied. DNA is negatively charged, so it migrates toward the positive electrode; the gel acts as a sieve, so smaller fragments move farther, giving a size-sorted band pattern.
3. Combined DNA Index System — a DNA database (not "Combined Official DNA Identification System").

Part D:

1. Any two: peat is darker (more organic matter); peat is more acidic (loam is near neutral); peat holds far more water; peat is made of partly decomposed plant material, loam is a sand/silt/clay mix.
2. Red = iron oxides (well-drained, oxidized). Gray = waterlogged / oxygen-poor (reduced iron). Pale/white = carbonates, salts, lime, or heavy leaching.
3. **Sandy clay loam** (55 sand / 20 silt / 25 clay).
4. A soil comparison (texture, color, pH, minerals) can put a sample in a group and strongly link or exclude a suspect, but many locations share the same soil profile, so it rarely proves one exact spot.

Part E:

1. Hole 1 came first — a crack stops when it hits an existing crack, so hole 1's cracks were already there when hole 2's cracks reached them.
2. From the outside — the cone-shaped hole widens in the direction the bullet traveled (narrow entry, wide exit).
3. The area of convergence. To get the height (area of origin) you also need the impact angle of each stain.

Part F:

1. Calcium carbonate (chalk / limestone) — insoluble + fizzes with acid; CaCO_3 .
2. Isopropyl (rubbing) alcohol — alcohol smell + fast evaporation + flammable; $\text{C}_3\text{H}_8\text{O}$.
3. Zinc — non-magnetic, density ~7.14, steady HCl reaction, atomic number 30.
4. Polypropylene (PP) — density ~0.90–0.92, so it floats in water and in oil but sinks in isopropyl; slow drip burn with a sweet/diesel smell. (It's the least dense of the common plastics.)

Part G — model answer:

- **Culprit: Morgan.** Motive: had keys and access; no alibi. (Opportunity is the main driver; a stated grudge isn't needed.)
- Evidence:
 - The glass fracture shows the force came **from inside the room** — consistent with someone already inside (staff), not someone breaking in from outside.
 - **Navy wool fiber** on the latch matches Morgan's navy wool coat (Riley wears white cotton; Casey wears polyester).
 - **White powder that fizzes but only partly dissolves** = a carbonate that isn't baking soda → calcium carbonate / garden lime; Morgan keeps garden lime by the loading dock.
 - **Sandy-loam soil** by the case matches the loading-dock area Morgan works in.
 - **Loop fingerprint** on the latch is consistent with Morgan (loop/loop) and **excludes Casey** (arch/tented arch).
- Timeline: Morgan has no alibi for the fundraiser; Riley was in the kitchen with three witnesses; Casey arrived and left with a group.
- Clear the others:
 - **Riley** — a white apron doesn't match the navy wool fiber, no soil or powder link, and a solid witnessed alibi.
 - **Casey** — has a motive (grudge), but **no** matching physical evidence (polyester not wool, no powder, no soil), the fingerprint excludes Casey, and Casey was with a group. Motive without evidence is not proof.